PROJECT SYNOPSIS

ON

AI DRIVEN AUTONOMOUS TRADING AGENT

Submitted to the Department of Computer Science and Engineering
in Partial Fulfillment of the Requirements
for the Degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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JSS ACADEMY OF TECHNICAL EDUCATION,NOIDA

DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW
(Formerly Uttar Pradesh Technical University, Lucknow)
SEPTEMBER, 2026

VISION AND MISSION

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JSS Academy of Technical Education Noida aims to become an Institution of excellence in imparting quality Outcome Based Education that empowers the young generation with Knowledge, Skills, Research, Aptitude and Ethical values to solve Contemporary Challenging Problems.

MISSION OF THE INSTITUTE

1. Develop a platform for achieving globally acceptable level of intellectual acumen and technological competence.
2. Create an inspiring ambience that raises the motivation level for conducting quality research.
3. Provide an environment for acquiring ethical values and positive attitude.

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"To spark the imagination of the Computer Science Engineers with values,skills and creativity to solve the real-world problems."

MISSION OF THE DEPARTMENT

1. To inculcate creative thinking and problem-solving skills through effective teaching, learning and research.
2. To empower professionals with core competency in the field of Computer Science and Engineering.
3. To foster independent and lifelong learning with ethical and social responsibilities.

PROGRAM OUTCOMES(POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify,formulate,review research literature,and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts,and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and teamwork: Function effectively as an individual,and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation,make effective presentations,and give and

receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work,as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for,and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OUTCOMES (PEOs)

PEO1: To apply computational skills necessary to analyze, formulate and solve engineering problems.

PEO2: To establish a entrepreneurs,and work in interdisciplinary research and development organizations as an individual or in a team.

PEO3: To inculcate ethical values and leadership qualities in students to have a successful career.

PEO4: To develop analytical thinking that helps them to comprehend and solve real-world problems and inherit the attitude of lifelong learning for pursuing higher education.

PROGRAM SPECIFIC OUTCOMES(PSOs)

PSO1: Acquiring in depth knowledge of theoretical foundations and issues in Computer Science to induce learning abilities for developing computational skills.

PSO2: Ability to analyse, design, develop, test and manage complex software system and applications using advanced tools and techniques.

Course Outcomes (COs)

C410.1: Identify, formulate, design, and analyze a research-based/web-based problem.

C410.2: Communicate effectively in verbal and written form.

C410.3: Apply appropriate computing and engineering skills for obtaining solutions to the formulated problem within a stipulated time.

C410.4: Work effectively as a part of a team in multidisciplinary areas.

C410.5: Consolidate the final outcome in the form of a publication.

CO-PO-PSO Mapping

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**
C410.1	3	3	3	3	2	3	3	3	3	3	2	3	3	3
C410.2	2	2	2	2	2	2	2	2	2	3	2	3	2	3
C410.3	3	3	3	3	3	3	2	3	3	3	3	3	3	3
C410.4	3	3	3	3	2	3	2	3	3	3	3	3	3	3
C410.5	3	3	3	3	3	3	3	3	3	3	3	3	3	3
C410	**2.80**	**2.80**	**2.80**	**2.80**	**2.40**	**2.80**	**2.40**	**2.80**	**2.80**	**3.00**	**2.60**	**3.00**	**2.80**	**3.00**

DECLARATION

We hereby declare that this synopsis submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that project synopsis report entitled “AI DRIVEN AUTONOMOUS TRADING AGENT” which is submitted by Soha , Punya & Pranjal in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science and Engineering of Dr. APJ Abdul Kalam Technical University,Uttar Pradesh, Lucknow is a record of the candidate’s own work carried out by him/her under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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AI DRIVEN AUTONOMOUS TRADING AGENT

ABSTRACT

Cryptocurrency trading is highly dynamic and requires continuous analysis of market conditions, timely decision-making and effective risk management. Conventional automated trading systems generally rely on predefined technical indicators and fixed trading rules, which can limit their ability to adapt to changing market conditions. The proposed AI Autonomous Trading System addresses these limitations by integrating technical analysis, Large Language Model (LLM)-based reasoning, risk management and automated trade execution into a single intelligent framework.

The system collects market data through APIs such as Binance and Lighter and processes OHLCV data along with indicators such as Exponential Moving Average (EMA) and Moving Average Convergence Divergence (MACD). Unlike systems that depend solely on predefined trading rules, our system employs an AI decision agent using an LLM through the Groq API. The agent considers multiple sources of context, including current market conditions, portfolio state, and previous trading history which is to generate informed trading decisions. A dedicated risk management layer evaluates these decisions before execution, reducing the possibility of uncontrolled or unsuitable trades. All approved trades are executed through the Lighter Testnet, while PostgreSQL and Prisma provide persistent storage of trading history, portfolio information and audit records.

The key advantage of the proposed system is its end-to-end integration of market analysis, contextual AI decision-making, risk evaluation, automated execution, and trade auditing within one system. This reduces dependence on isolated rule-based strategies and provides a more adaptive and structured approach to autonomous trading. The modular architecture also allows individual components to be improved or extended without redesigning the complete system. Thus, the proposed system provides a comprehensive and scalable framework for intelligent cryptocurrency trading and establishes a practical foundation for developing more adaptive autonomous trading solutions.

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LIST OF SYMBOLS/ABBREVIATIONS

AI Artificial Intelligence

API Application Programming Interface

EMA Exponential Moving Average

LLM Large Language Model

MACD Moving Average Convergence Divergence

OHLCV Open, High, Low, Close, Volume

SDK Software Development Kit

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CHAPTER 1

INTRODUCTION

Cryptocurrency markets operate continuously across global exchanges, generating price signals twenty-four hours a day, seven days a week. The velocity and complexity of these markets make sustained manual monitoring infeasible, as profitable opportunities can emerge and vanish within seconds far beyond the reaction capacity of a human trader.

Cognitive fatigue, slow reaction time and the inability to process multiple market signals simultaneously makes human structurally disadvantaged for continuous, high-frequency market monitoring and decision execution. Conventional automated systems attempt to address this gap, but they typically depend on fixed, rule-based logic that struggles to adapt when market conditions change. Generative AI and Large Language Models (LLMs) offer a fundamentally different approach. Rather than following static rules, an LLM can reason holistically over complex, multi-factor market contexts, integrating technical indicators, portfolio state, and historical behaviour into a single coherent decision. This enables adaptive, context-aware decisionmaking that surpasses the capability of static rule-based systems.

Our project addresses this opportunity by developing an autonomous, AI-driven trading agent that collects live market data , analyzes it and calculates the parameters and account information together, reasons over this context using a Large Language Model, and automatically executes trading decisions on chain, without manual intervention. The system operates on the ETH-USDC market on the Lighter Protocol testnet, and is made so to respond instantly to market signals while operating continuously without downtime or fatigue and integrate multiple technical indicators into easily explainable trading decisions.

By automating the entire loop of market analysis, decision-making, and trade execution, the project eliminates the structural disadvantages of manual trading and the rigidity of rule-based bots, while providing a transparent, auditable record of every decision the agent makes.

CHAPTER 2

MOTIVATION

The motivation for this project stems from the broad gap between the speed at which cryptocurrency markets are moving and the speed at which a human trader or even a conventional rule based bot can reliably respond. Markets such as ETH-USDC trade continuously and meaningful price movements can occur within seconds making disciplined, round the clock monitoring practically impossible for an individual and rigid for a static algorithm.

Existing automated trading tools generally fall into two categories: first is , rule based systems that encode a fixed set of conditions (for example, moving-average crossovers) and second is , purely predictive machine learning models that forecast price direction without an integrated decision and execution loop. Neither approach adapts well to shifting market trends, nor do they typically provide a transparent, human-readable rationale for the actions they take. This makes it difficult for a trader to trust, audit or improve such systems over time.

Recent advances in Large Language Models make it possible to reason over rich, multifactor context, such as technical indicators, portfolio exposure, and recent trading history, in a manner closer to how an experienced trader would weigh several signals before acting. This motivated the idea of building a trading agent in which an LLM sits at the centre of the decision loop, supported by conventional technical analysis, risk constraints and a persistent audit trail, rather than a system driven purely by static rules or by prediction alone.

Deploying such a system on the Lighter Protocol testnet, rather than directly on live markets, is further motivated by the need to study the complete behaviour of an autonomous LLM-driven trading pipeline, its decision quality, its response to changing conditions, and its risk-management discipline, in a safe and controlled environment before any consideration of real capital. This project is therefore driven by both a practical need (continuous, fatigue

free market monitoring) and a research interest (evaluating LLMbased reasoning as a viable decision-making mechanism for algorithmic trading).

CHAPTER 3

OBJECTIVES

The primary objective of the proposed idea is to develop an intelligent, automated and context aware trading system that is capable of continuously collecting and analyzing real-time cryptocurrency market data, making informed trading decisions, executing trades autonomously while maintaining a comprehensive record of its activities. The proposed system integrates real time market data collection, technical analysis, Large Language Model (LLM)-based decision-making, risk aware trade execution, persistent audit logging and API-based monitoring into a unified automated pipeline.

The proposed project aims to achieve the following primary deliverables:

- **Research Paper**
  - To conduct research on the application of Artificial Intelligence, Large Language Models (LLMs), and autonomous agents in cryptocurrency trading.
  - To analyze, document, and present the methodology, system design, implementation, and experimental findings in the form of a research paper.
- **AI-Driven Autonomous Trading Agent**
  - To design and develop an AI-Driven Autonomous Trading Agent capable of collecting market data, analyzing market conditions, generating trading decisions, and executing trades autonomously.
  - To implement and evaluate the proposed system in a controlled cryptocurrency trading environment using the Lighter Protocol Testnet.

CHAPTER 4

SCOPE OF THE PROJECT

The scope of the proposed AI-Driven Autonomous Trading Agent includes the following major functionalities and research areas:

- **Real-Time Market Data Acquisition**
  - Collection of live OHLCV (Open, High, Low, Close, Volume) market data from the Lighter Protocol Testnet.
  - Continuous updating and processing of market information for analysis and decision-making.
  - Preparation of structured datasets suitable for technical analysis and AI-based reasoning.
- **Technical Analysis and Market Trend Identification**
  - Computation of technical indicators such as Exponential Moving Average (EMA) and Moving Average Convergence Divergence (MACD).
  - Analysis of short-term and long-term market trends using multiple timeframes.
  - Generation of analytical insights from processed market data.
- **LLM-Based Trading Decision Generation**
  - Utilization of Large Language Models (LLMs) for context-aware trading decision making.
  - Incorporation of market conditions, technical indicators, portfolio state, and historical context into the decision process.
  - Generation of trading actions such as LONG, SHORT, CLOSE, and HOLD.

- **Autonomous Trade Execution**
  - Automated execution of trading actions through the Lighter Protocol Testnet.
  - Secure local transaction signing and authenticated trade submission.
  - End-to-end integration from decision generation to trade execution.
- **Risk Management and Trade Validation**
  - Evaluation of proposed trading decisions against predefined constraints and portfolio rules.
  - Control of position exposure and execution limits before order placement.
  - Support for responsible and risk-aware autonomous trading.
- **Audit Logging and Data Persistence**
  - Storage of agent activities, trading decisions, execution records, and portfolio information.
  - Maintenance of a complete audit trail for transparency and traceability.
  - Support for historical analysis and performance evaluation.
- **Monitoring and Observability**
  - Provision of API-based monitoring endpoints.
  - Real-time visibility into trading activity, portfolio state, execution status, and system health.
  - Facilitation of analysis without direct interaction with internal components.
- **Backtesting and Strategy Evaluation**
  - Analysis of trading strategies using historical market data.
  - Evaluation through metrics such as profitability, trade frequency, win rate, and drawdown.
  - Comparison of different strategy configurations in a controlled environment.
- **Educational and Research Applications**

  - Demonstration of practical applications of Data Science, Artificial Intelligence, and Financial Analytics.
  - Study of autonomous agents, technical analysis, time-series data processing, and algorithmic trading concepts.
  - Use as a research and experimentation platform for intelligent financial systems.

- **Future Extension Possibilities**
  - Support for multiple cryptocurrencies and portfolio-level analysis.
  - Integration of financial news, sentiment analysis, and additional market indicators.
  - Exploration of advanced Machine Learning and Reinforcement Learning techniques.
  - Enhancement through cloud deployment, scalability improvements, and advanced monitoring systems.

CHAPTER 5

RELATED WORK

5.1 Existing Approach

Traditional cryptocurrency trading systems primarily rely on predefined technical indicators, fixed trading rules or manually configured strategies. These systems generally analyze historical and real-time market data using indicators such as moving averages, Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD) and other technical analysis techniques.

Although rule-based systems can provide consistent execution, they have limited ability to interpret complex market conditions or adapt their decision-making according to changing market contexts. In addition, many existing automated trading systems focus primarily on generating trading signals rather than integrating market analysis, portfolio state, risk management, and trade execution into a single autonomous system.

Recent research has explored the use of Large Language Models (LLMs) and AI agents in financial applications. LLM-based systems provide the ability to process contextual information and generate decisions based on multiple sources of information. However, integrating LLM reasoning with real-time financial data, risk constraints, and actual trade execution remains a challenging problem.

Table 5.1: Existing Approach vs Limitations

Existing Approach	**Limitations**
Rule-based trading systems	Depend on predefined rules and have limited adaptability to changing market conditions.
Technical indicator-based systems	Primarily depend on numerical indicators and may not consider broader market context.
Automated trading bots	Often focus on execution and predefined strategies rather than context-aware decision-making.
LLM-based financial agents	Require reliable integration with real-time data, risk management, and secure trade execution.

5.2 Proposed Approach

The proposed AI-Driven Autonomous Trading Agent combines real-time cryptocurrency market analysis, technical indicators, Large Language Model-based reasoning, risk management, autonomous trade execution, and persistent logging into a unified system.

The system continuously collects market data and account information, including OHLCV data, portfolio state, open positions, and trading history. Technical indicators such as Exponential Moving Average (EMA) and Moving Average Convergence Divergence (MACD) are calculated from the market data to provide additional information about market trends and momentum.

The processed market information is provided to an LLM-based decision agent. The agent considers the current market context, portfolio state, and previous trading activity before generating a trading decision. The possible trading actions include LONG, SHORT, and CLOSE, depending on the current market and portfolio conditions.

Before a trade is executed, the decision passes through a risk management layer that applies predefined constraints. Approved actions are then executed through the Lighter Protocol Testnet. The system also maintains persistent records of agent invocations, tool calls,

trading decisions, execution details, and portfolio information using the database layer.

This integrated approach aims to provide an autonomous trading pipeline in which data collection, analysis, decision-making, risk management, execution, and monitoring operate as interconnected components rather than as independent modules.

5.3 Future Scope

The proposed system can be further extended to improve its capabilities and applicability. Future development may include support for additional cryptocurrency markets and trading pairs, more advanced technical indicators, and additional market data sources.

The decision-making component can also be extended with more advanced LLM models and additional contextual information. Historical trading data can be used for more extensive evaluation and strategy analysis.

Further improvements may include advanced risk-management techniques, portfolio optimization, improved monitoring dashboards, and additional performance metrics. The system could also be evaluated under different market conditions and extended from the testnet environment to other controlled trading environments after appropriate testing and validation.

CHAPTER 6

HARDWARE AND SOFTWARE REQUIREMENTS

6.1 Software Requirements

The software requirements define the technologies, frameworks, libraries, and development tools required for implementing the proposed AI-Driven Autonomous Trading Agent.

Table 6.1: Software Requirements

Software	**Requirement / Purpose**
Operating System	Windows / Linux / macOS for development and execution.
Programming Environment	Node.js-based development environment for implementing the autonomous trading system.
Database	PostgreSQL for persistent storage of trading history, portfolio information, and audit logs.
ORM	Prisma for database access and management.
AI / LLM Service	Groq API for LLM-based trading decision-making.
Trading Platform	Lighter Protocol Testnet for controlled trade execution and testing.
API Services	Market data and trading APIs for retrieving market information and executing trades.

6.2 Hardware Requirements

The hardware requirements for the proposed system are relatively moderate because the system uses external APIs and an LLM service rather than requiring local model training.

Table 6.2: Hardware Requirements

Component	**Requirement**
Processor	Intel Core i5 or equivalent processor.
RAM	Minimum 8 GB RAM.
Storage	Minimum 256 GB available storage.
Internet Connection	Stable internet connection for real-time market data, API communication, LLM requests, and testnet interaction.
System	A standard desktop or laptop computer capable of running the required development environment.

6.3 External Services and APIs

The system depends on external services and APIs for market data, LLM-based decision-making, and blockchain-based trade execution.

Table 6.3: External Services and APIs

Service / API	**Purpose**
Lighter Protocol	Provides the testnet environment for cryptocurrency trading and trade execution.
Groq API	Provides access to the Large Language Model used for context-aware trading decisions.
Market Data APIs	Provide real-time cryptocurrency market information including OHLCV data.

6.4 Major Project Dependencies

The major dependencies required for the implementation of the project include the software libraries, database components, APIs, and development tools used by the system.

Table 6.4: Major Project Dependencies

Dependency	**Purpose**
Node.js	Runtime environment for executing the application.
Prisma	Database ORM used for interacting with the persistent database.
PostgreSQL	Database for storing trading history, audit information, and portfolio-related data.
Groq API	LLM service used by the AI decision-making component.
Lighter SDK	Provides interaction with the Lighter Protocol for trading operations.

CHAPTER 7

PREDICTED TIMELINE

The development of the AI-Driven Autonomous Trading Agent is planned through multiple phases including requirement analysis, system design, implementation, integration, testing, and documentation. The estimated timeline for completing the project is illustrated in Figure 7.1.

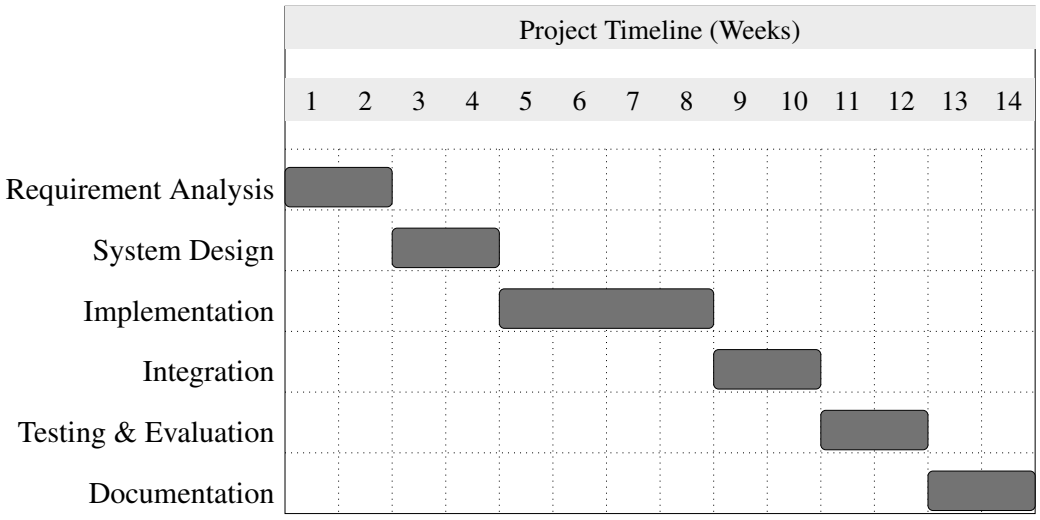


Figure 7.1: Predicted Gantt Chart for the AI-Driven Autonomous Trading Agent

Timeline Description:

The project is expected to be completed within fourteen weeks. The initial phase focuses on requirement analysis and understanding the trading environment. This is followed by system design, where the architecture, database schema, and component interactions are defined. The implementation phase constitutes the core development effort, including market data processing, technical indicator generation, AI-based decision-making, and risk management modules. Subsequently, integration with external services such as the Lighter Protocol Testnet, Large Language Model (LLM) APIs, and database systems is performed. The devel-

oped system is then evaluated through comprehensive testing to verify trading performance, reliability, and safety mechanisms. Finally, documentation and report preparation are completed for project submission.

CHAPTER 8

CONCLUSION

8.1 Conclusion

The proposed AI Autonomous Trading Project presents an integrated framework for automated cryptocurrency trading by combining real-time market data processing, technical analysis, artificial intelligence-based decision making, risk management, and automated trade execution. The system is designed to minimize the need for continuous manual monitoring while providing a structured and auditable approach to autonomous trading.

The system collects market and account information through external APIs and processes OHLCV data to generate technical indicators such as Exponential Moving Average (EMA) and Moving Average Convergence Divergence (MACD). The processed market information, together with portfolio information and previous trading history, is provided to the AI decision agent to support context-aware trading decisions.

The integration of a Large Language Model (LLM) through the Groq API enables the system to interpret the available market context and generate trading decisions. These decisions are subsequently evaluated through the risk management layer before being forwarded to the trading execution module. The execution component is connected to the Lighter Testnet, providing a controlled environment for testing the proposed autonomous trading workflow.

The system also incorporates persistent audit logging using PostgreSQL and Prisma. This enables trading decisions, execution details, and relevant system information to be recorded for monitoring, analysis, and future evaluation. The modular architecture allows the individual components of the system to work together as a complete automated trading pipeline.

Overall, the proposed project demonstrates the feasibility of integrating market data analysis, AI-based decision making, risk management, automated execution, and audit logging

into a unified autonomous trading system. The project provides a foundation for further experimentation and development in the field of AI-assisted algorithmic cryptocurrency trading.

8.2 Future Scope

The proposed system can be further enhanced by incorporating additional data sources, analytical techniques, and evaluation mechanisms. The major areas of future development include:

- **Extended Technical Indicators:** The system can be enhanced by incorporating additional technical indicators such as Relative Strength Index (RSI), Bollinger Bands, Average True Range (ATR), and Stochastic Oscillator. These indicators can provide the AI decision agent with a broader representation of market trends, momentum, volatility, and potential entry or exit conditions.
- **Sentiment Integration:** Market sentiment can be incorporated as an additional source of information for the AI decision agent. Sentiment data obtained from sources such as news, social media, and other market-related information can be processed to identify positive, negative, or neutral market sentiment and combine it with technical market signals.
- **Backtesting Framework:** A dedicated backtesting framework can be developed to evaluate the trading logic and AI-generated decisions using historical market data. This would allow different strategies, indicator combinations, and risk-management parameters to be tested before deployment in a live or testnet environment.
- **WebSocket-Based Market Data Feed:** The current API-based market data pipeline can be extended using WebSocket feeds to support continuous and low-latency delivery of market updates. This can improve the responsiveness of the system and enable more timely processing of rapidly changing market conditions.

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APPENDIX A

PLAGIARISM REPORT

A.1 Plagiarism Report

A.2 AI Content Report